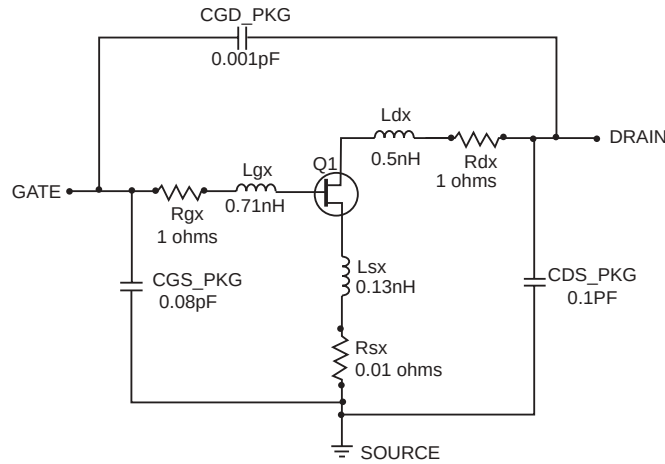


SCHEMATIC



FET NONLINEAR MODEL PARAMETERS ⁽¹⁾

Parameters	Q1	Parameters	Q1
VTO	-2.24	RG	8
VTOSC	0	RD	0.5
ALPHA	8	RS	6
BETA	0.055	RGMET	0
TQGAMMA	0.04	TNOM	27
TQGAMMAAC	0.05	XTI	3
Q	1.5	EG	1.43
TQDELTA	0.25	VTOTC	0
VBI	1	BETATCE	0
IS	1e-14	FFE	1
N	1.3	FNC ⁽²⁾	150e-6
RIS	0	R	0.5
RID	0	P	1
TAU	3e-12	C	0.9
CDS	0.19e-12		
RDB	250		
CBS	1e-9		
CGS	0.92e-12		
CGD	0.05e-12		
DELTA1	0.3		
DELTA2	0.3		
FC	0.5		
VBR	Infinity		

(1) ADS TOM Model

(2) To simulate phase noise using AF/KF: AF = 1.5 KF = 2e-10

Life Support Applications

These NEC products are not intended for use in life support devices, appliances, or systems where the malfunction of these products can reasonably be expected to result in personal injury. The customers of CEL using or selling these products for use in such applications do so at their own risk and agree to fully indemnify CEL for all damages resulting from such improper use or sale.

UNITS

Parameter	Units
time	seconds
capacitance	farads
inductance	henries
resistance	ohms
voltage	volts
current	amps

MODEL RANGE

Frequency: 0.1 to 18 GHz
 Bias: $V_{DS} = 2\text{ V to }4\text{ V}$, $I_D = 10\text{ mA to }40\text{ mA}$
 Date: 02/2002

EXCLUSIVE NORTH AMERICAN AGENT FOR NEC RF, MICROWAVE & OPTOELECTRONIC SEMICONDUCTORS

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02/13/2002